TASK WK1-PM1

Setup a Cybersecurity testing lab environment on your Laptop/PC with below detail:

- Use Virtualbox as base (latest recommended version)
- Setup Kali Linux as attacking/hacker machine
- Setup the network in subnet 10.0.0.0/24
- Use NATNetwork with 10.0.0.0/24
- The clipboard & file drag/drop should be enabled in Virtual Machine settings
- Shared folders should be enabled with /downloads folder shared from host machine
- Kali Linux IP Address should be 10.0.0.2/24
- Kali Linux should have full Internet access

Lab Setup (for Cyber Security & Ethical Hacking practice)

Network Diagram & architecture

Main OS
(Windows10)
IP: x.x.x.x

VirtualBox

Network Settings on all VM's:

NATNetwork (custom)

Range: 10.0.0.2-99 /24

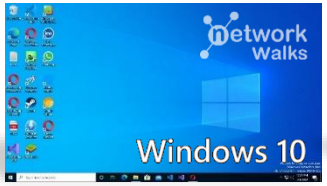


Manual IP: 10.0.0.2/24 (NATNetwork)

Manual IP: 10.0.0.11 /24
(NATNetwork)



Manual IP: 10.0.0.10 /24
(NATNetwork)



Manual IP: 10.0.0.7 /24
(NATNetwork)



Follow networkwalks.com for detail
(Optional)

Manual IP: 10.0.0.16 /24
(NATNetwork)



(Optional)

Manual IP:
10.0.0.9 /24
(NATNetwork)



Android

PC/Laptop specs for this lab setup
(recommended but not a must)
- 8GB RAM or more
- 256GB SSD or more
- Core i3/i5 Processor (or any similar)

Lab Setup (for Cyber Security & Ethical Hacking practice)

SOLUTION

Follow the below steps & follow the video guide as explained by instructor in the LIVE class.

PHASE 1

Step 1. Download & install 7-zip: https://7-zip.org/download.html

Step 2. Download & install Virtualbox on your laptop/PC: https://virtualbox.org/wiki/Downloads

Step 3. Configure the network settings on your Virtualbox (create NATNetwork in 10.0.0.0/24)

Step 4. Download & import Kali Linux Virtual Machine in your Virtualbox: https://kali.org/get-kali

Step 5. Setup the IP configuration of Kali Linux

Step 6. Take snapshot of the VM

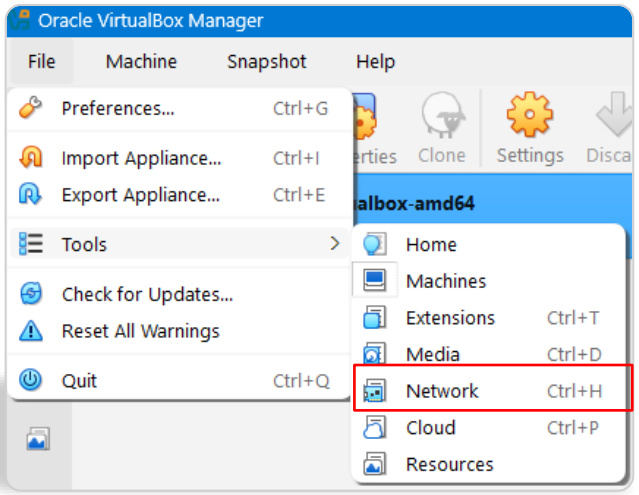
PHASE 2 (for future optional tasks)

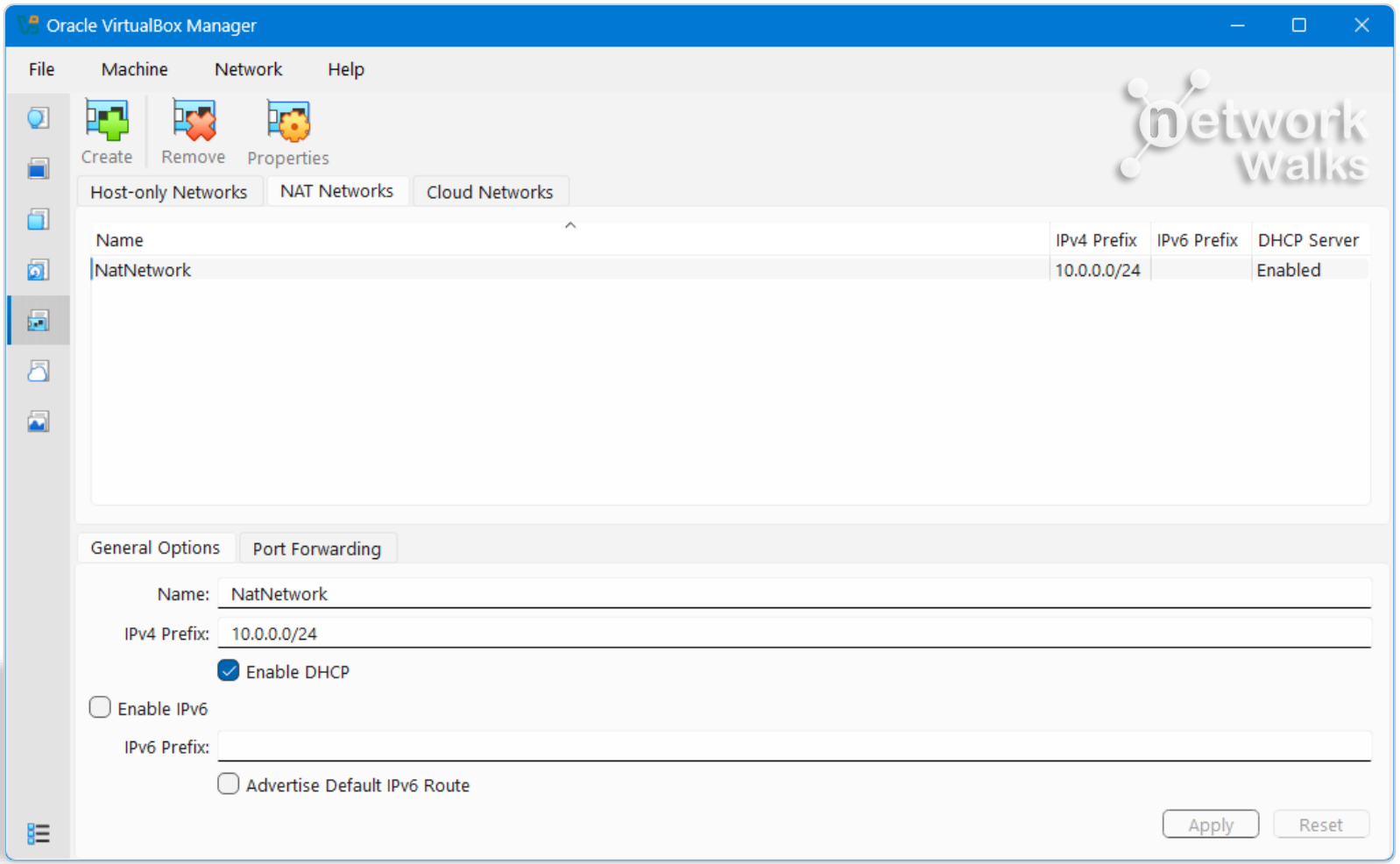
Step 7. Download & Install Windows 11/10/7/Android9x on your Virtualbox Virtual Machine, ping test all machines from each other & take snapshots.

**There might be further offline Virtual Machines for CTF practical Labs & challenges*

Lab Setup (for Cyber Security & Ethical Hacking practice)

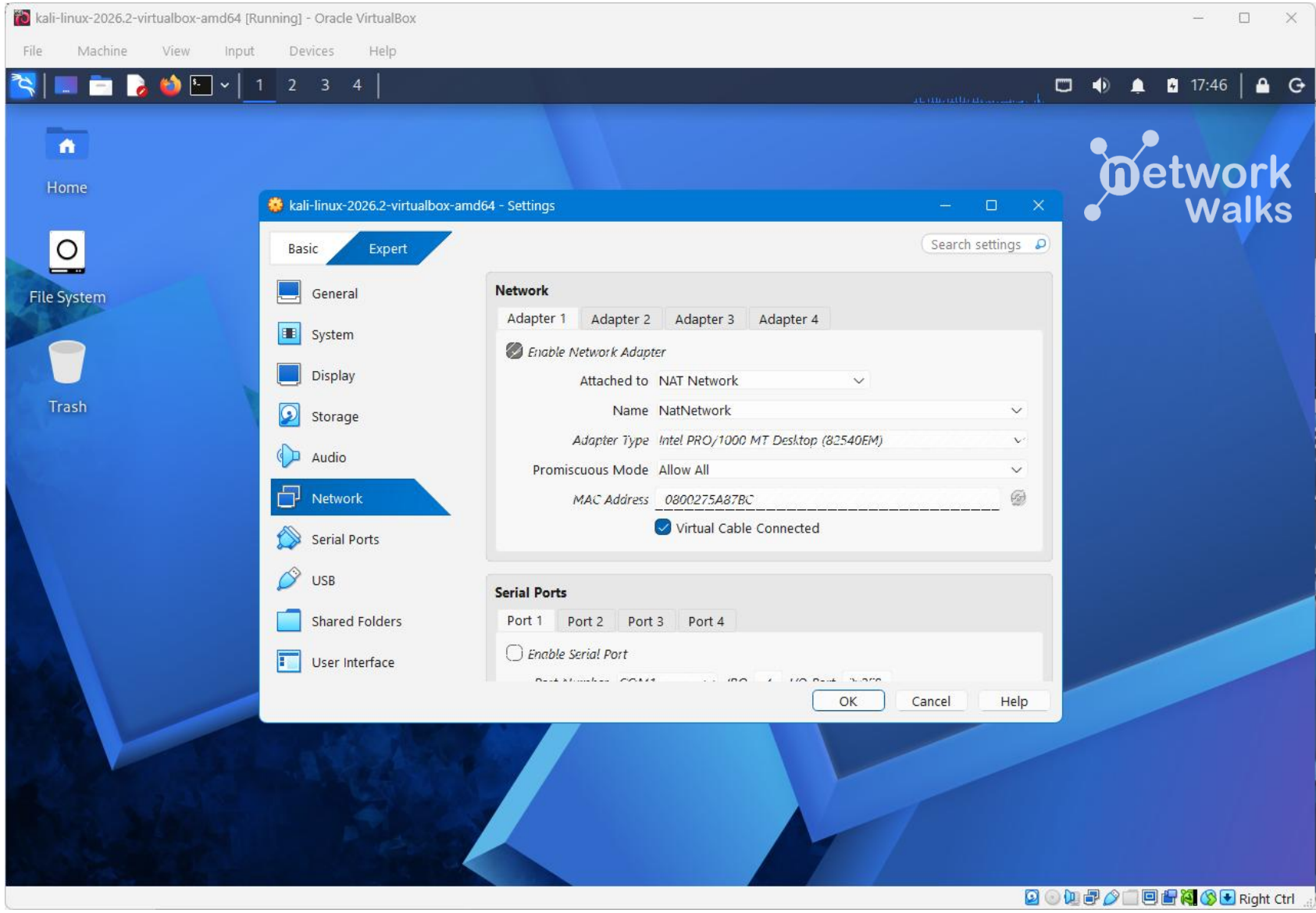
Virtualbox Settings - Network





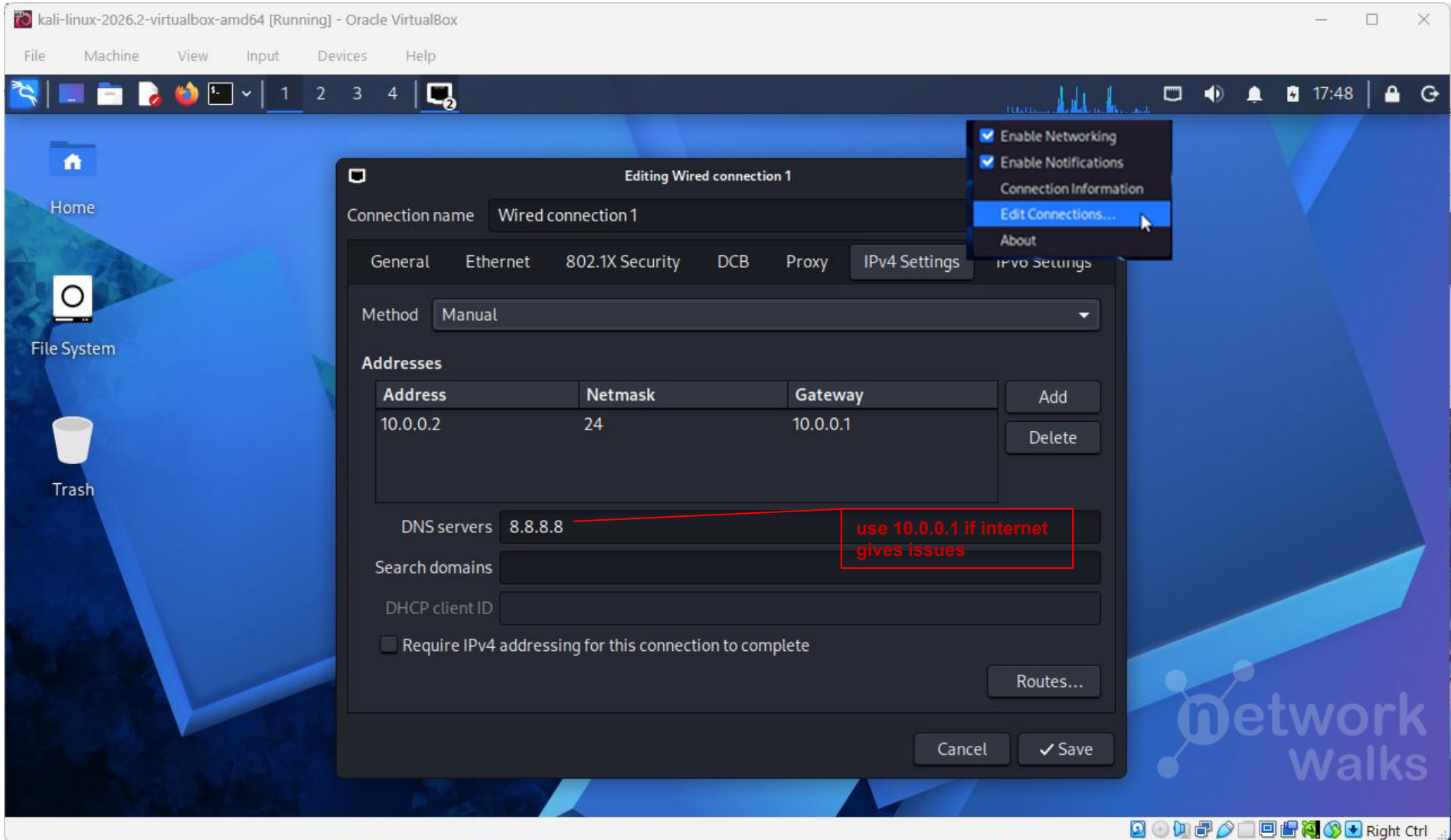
Lab Setup (for Cyber Security & Ethical Hacking practice)

Virtualbox Settings for Kali Linux VM



Lab Setup (for Cyber Security & Ethical Hacking practice)

Virtualbox Settings for Kali Linux VM



Lab Setup (for Cyber Security & Ethical Hacking practice)

What to do if your Kali VM has no Internet?

1. Check if you created NATNetwork properly
2. Check if all your network settings are correct
3. Check that no other VM on the same NAT Network is using 10.0.0.2
4. Run below 3 commands & restart your Kali Linux
5. Restart your Virtual machines & your main OS
6. Follow the comments of relevant Youtube video (various students have shared their issues & solutions that how they resolve their issues).
7. Write your issue in Youtube video comments with your correct name. So that instructor & community could help you to resolve quickly.

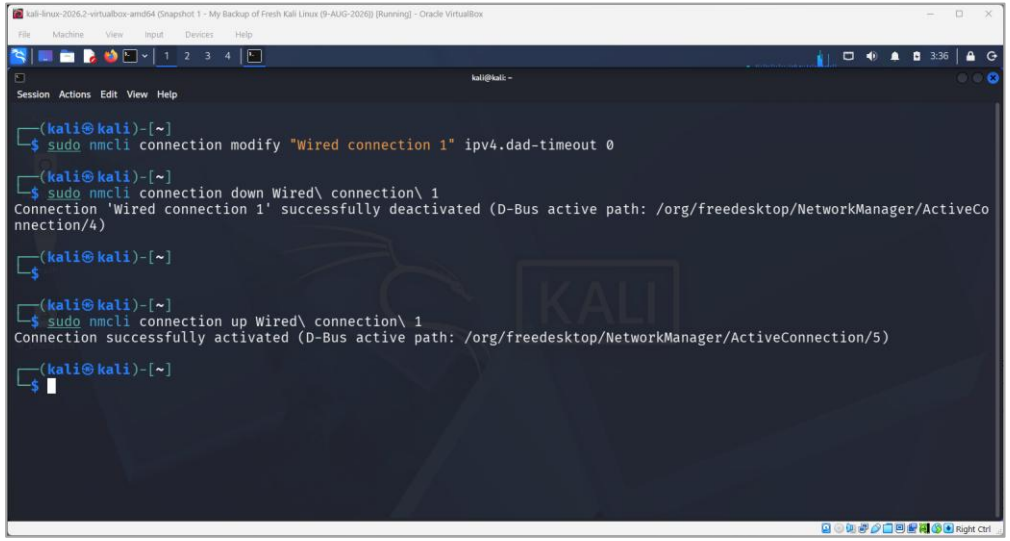
** It is a common issue in Virtualbox v7, Kali Linux 2026.1 or newer versions.*

Use below commands to fix if you face Internet connectivity issues (for Virtualbox v7, Kali Linux 2026.1 or higher versions):

sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0

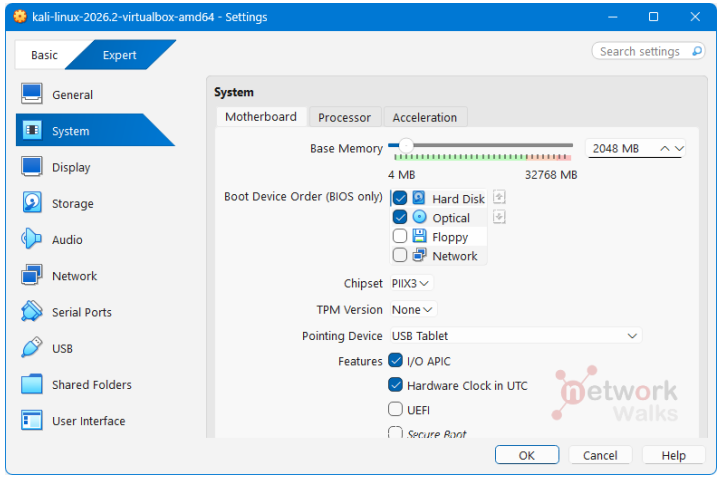
sudo nmcli connection down "Wired connection 1"

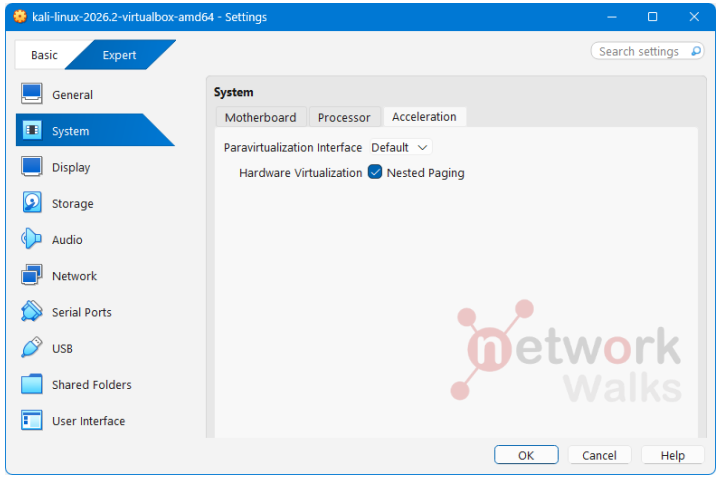
sudo nmcli connection up "Wired connection 1"

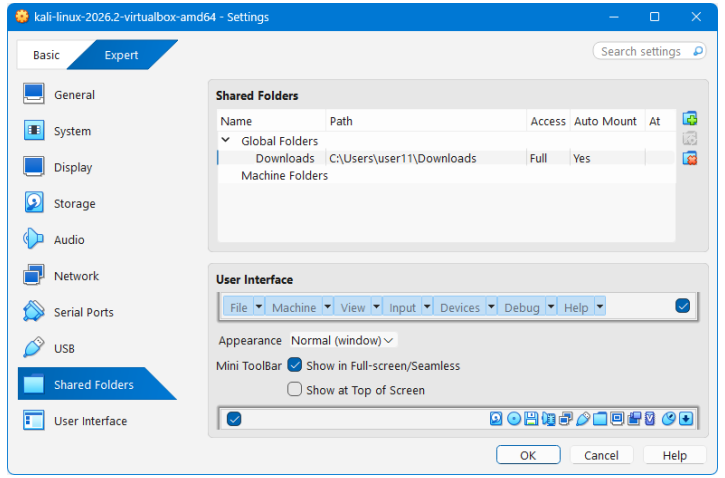


Lab Setup (for Cyber Security & Ethical Hacking practice)

Virtualbox Settings for Kali Linux VM

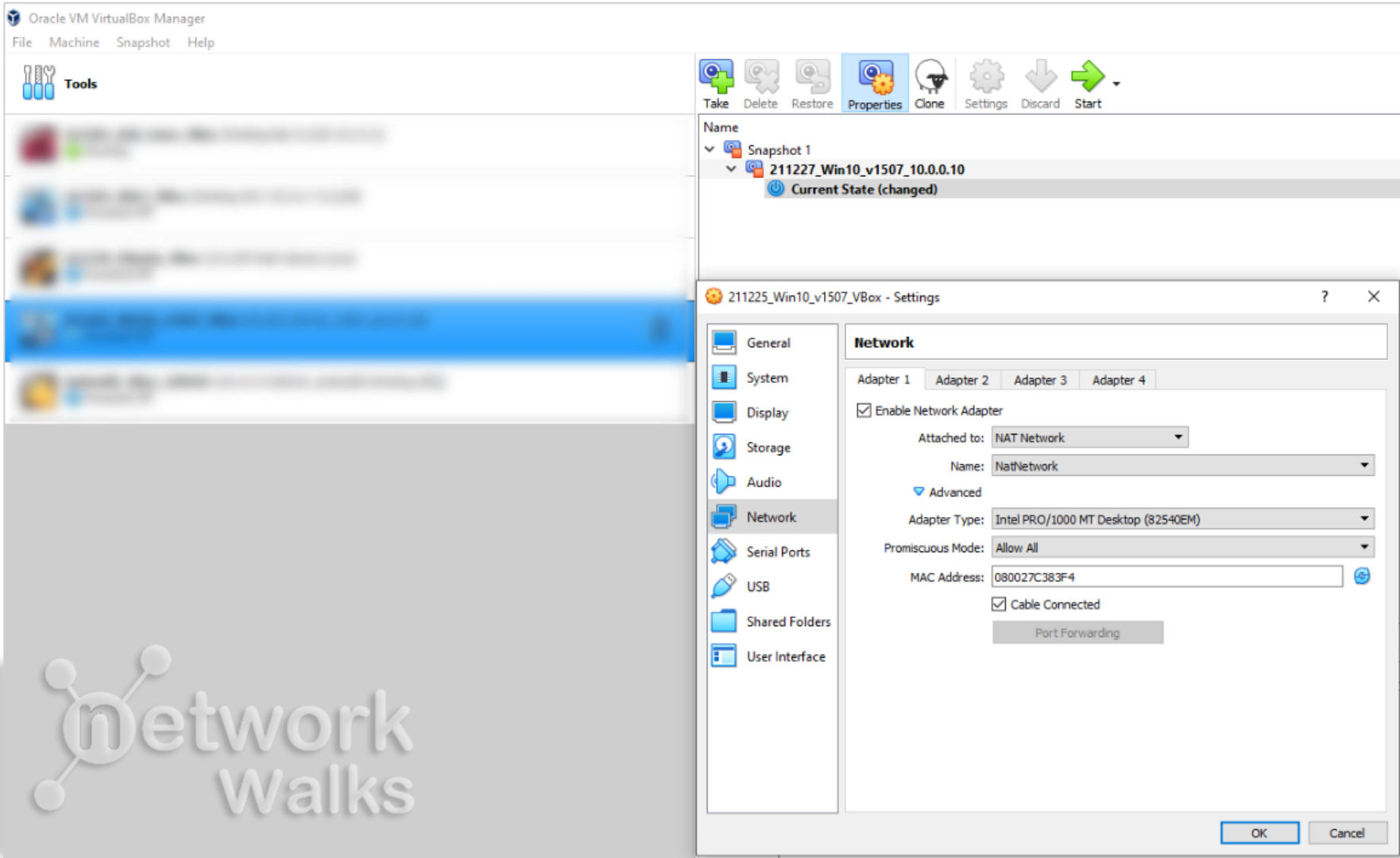






Lab Setup (for Cyber Security & Ethical Hacking practice)

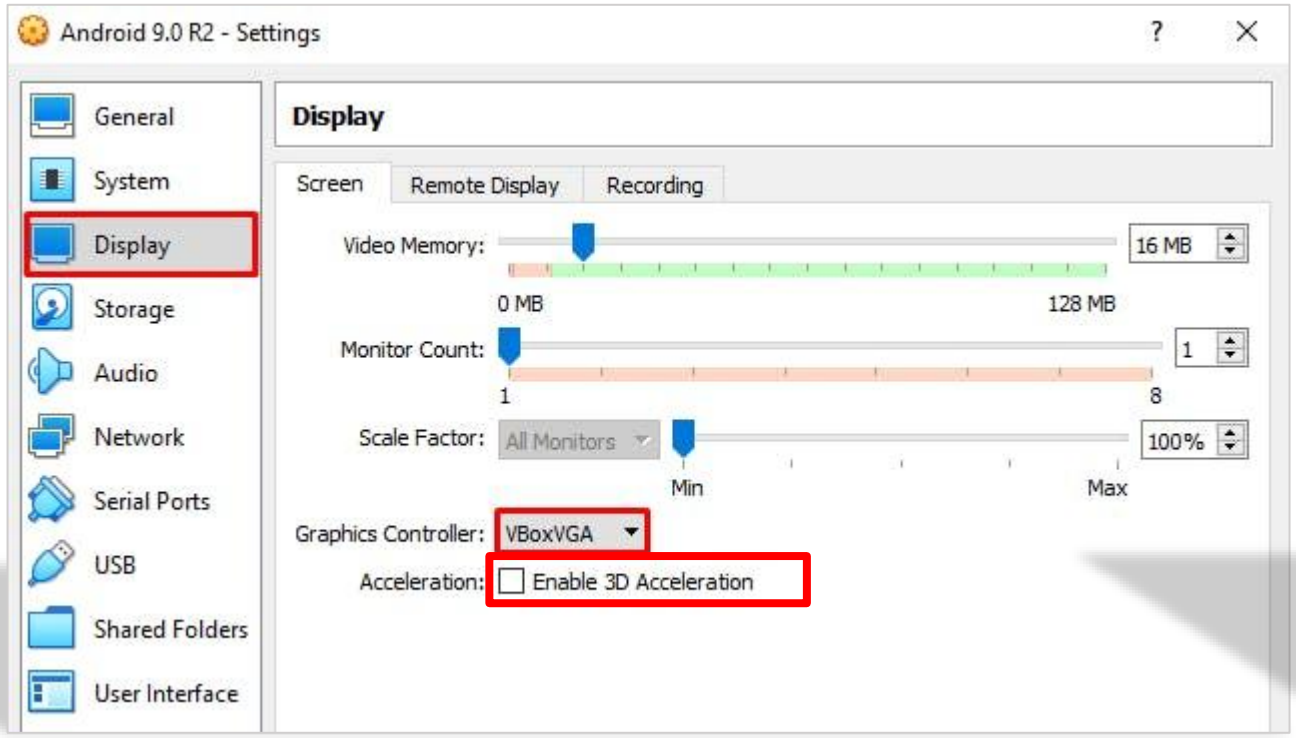
Virtualbox Settings for Windows10 VM

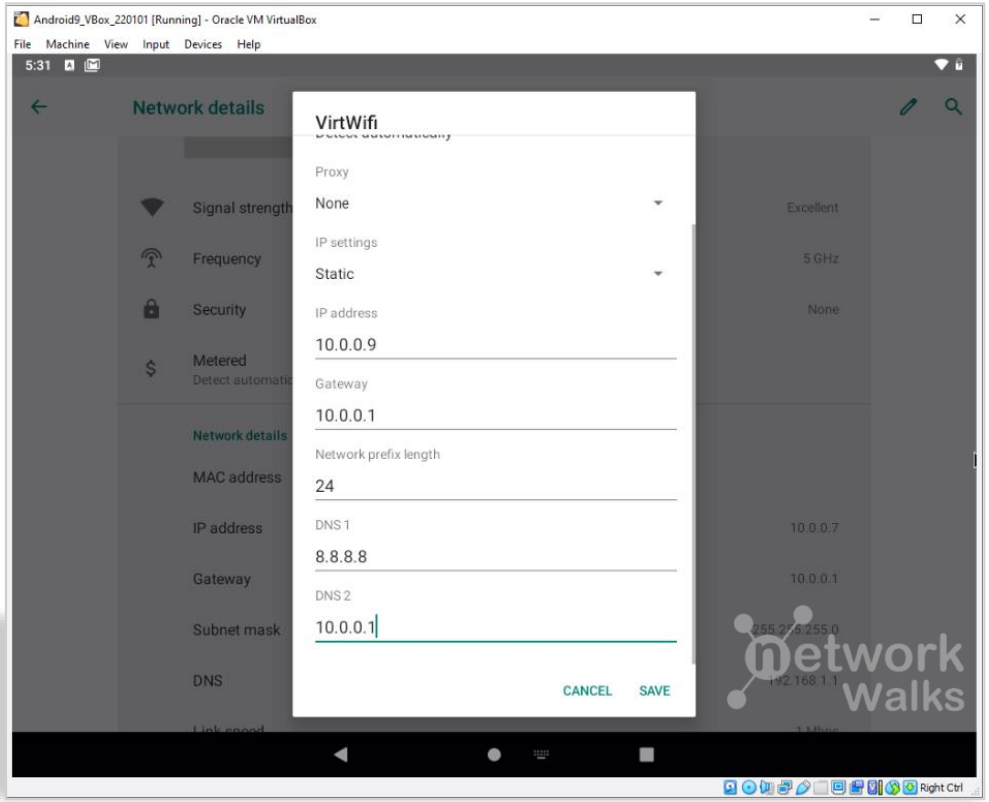


Network Walks

Lab Setup (for Cyber Security & Ethical Hacking practice)

Virtualbox Settings for Android VM





Thank You

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Your Technical Questions, comments & suggestions are always welcomed.



+277 66 2222 18 **+27 788 220 600**